

FUNCTIONS

WORKED OUT EXAMPLES

W.E.1: The function $f(x) = |px - q| + r|x|$, where

$x \in (-\infty, \infty)$ where $p > 0, q > 0, r > 0$ assume its minimum value only at one point, if

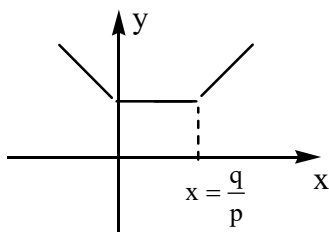
a) $p \neq q$ b) $r \neq q$ c) $r \neq p$ d) $p = q = r$

Key. c

Sol.
$$f(x) = \begin{cases} -px + q - rx; & x \leq 0 \\ -px + q + rx; & 0 < x \leq \frac{q}{p} \\ px - q + rx; & \frac{q}{p} < x \end{cases}$$

(i) for $r = p$,
$$f'(x) = \begin{cases} < 0; & \text{if } x < 0 \\ = 0; & \text{if } 0 < x < \frac{q}{p} \\ > 0; & \text{if } x > \frac{q}{p} \end{cases}$$

Here, we have infinitely many point for min in value for $f(x)$.



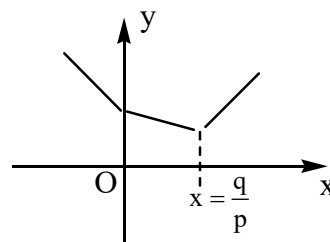
(ii) for $r < p$, $f'(x) < 0$; if $x \leq 0$

$$< 0; \text{ if } 0 < x \leq \frac{q}{p}$$

$$> 0; \text{ if } x > \frac{q}{p}$$

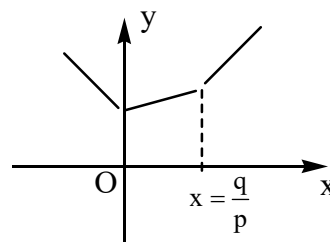
Here we have only one point of min. of $f(x)$ at

$$x = \frac{q}{p}$$



(iii) for $r > p$, $f'(x) < 0$; if $x \leq 0$
 > 0 ; if $x > 0$

from the graph, we have only one point of minimum at $x = 0$.



W.E-2: If $f(x) = x^2 + 2bx + 2c^2$ and

$g(x) = -x^2 - 2cx + b^2$ such that

$\min f(x) > \max g(x)$, then the relation between 'b' and 'c' is

a) no real value of b & c

b) $0 < c < b\sqrt{2}$ c) $|c| < |b|\sqrt{2}$

d) $|c| > b\sqrt{2}$

Key. d

Sol. We have min. $f(x) = 2c^2 - b^2$;

$$\max. g(x) = b^2 + c^2$$

$$f_{\min} > g_{\max} \Rightarrow 2c^2 - b^2 > b^2 + c^2$$

$$\Rightarrow |c| > |b|\sqrt{2}$$

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JE ADVANCED - VOL - III

W.E-3: X and Y are two sets and $f : X \rightarrow Y$. If

$$\{f(c) = y; c \in X, y \in Y\} \text{ and}$$

$\{f^{-1}(d) = x; d \in Y, x \in X\}$, then the true statement, is

a) $f(f^{-1}(b)) = b$ b) $f^{-1}(f(a)) = a$

c) $f(f^{-1}(b)) = b; b \subset y$

d) $f^{-1}(f(a)) = a; a \subset x$

Key. d

Sol. Given that X and Y are two sets and $f : X \rightarrow Y$,

$$\{f(c) = y; c \in X, y \in Y\} \text{ and}$$

$$\{f^{-1}(d) = x; d \in Y, x \in X\}$$

$$\text{since } f^{-1}(d) = x \Rightarrow f(x) = d$$

$$\text{Now if } a \in X \Rightarrow f(a) \in f(X) = d$$

$$\Rightarrow f^{-1}f(a) = a$$

$$f^{-1}f(a) = a, a \in X$$

W.E-4: If $f(x)$ satisfies the functional equation

$$x^2 \cdot f(x) + f(1-x) = 2x - x^4, \text{ then } f\left(\frac{1}{3}\right) =$$

a) $\frac{1}{3}$ b) $\frac{1}{9}$ c) $\frac{8}{9}$ d) $\frac{10}{9}$

Key. c

Sol. Replace 'x' by '1-x', we have

$$(1-x)^2 \cdot f(1-x) + f(x) = 2(1-x) - (1-x)^4$$

Now eliminating $f(1-x)$

$$\Rightarrow f(x) = \frac{(2x - x^4)(1-x)^2 - 2(1-x)(1-x)^4}{x^2(1-x)^2 - 1}$$

$$= 1 - x^2 \Rightarrow f\left(\frac{1}{3}\right) = \frac{8}{9}$$

W.E-5: Let $f : R \rightarrow R$ be a function such that for some

$$p > 0, f(x+p) = \frac{f(x)-5}{f(x)-3}, \forall x \in R, \text{ then}$$

the period of 'f' is

a) 2p b) 4p c) 3p d) 5p

Key. b

Sol. 3 does not belong to the range of 'f'

Now

$$f(x+2p) = f(x) \Rightarrow f(x+p+p) = f(x)$$

$$\begin{aligned} & \frac{f(x)-5}{f(x)-3} - 5 \\ \Rightarrow & \frac{f(x)-5}{f(x)-3} - 3 = \frac{-4f(x)+10}{-2f(x)+4} \end{aligned}$$

$$= \frac{2f(x)-5}{f(x)-2}$$

$$(f(x)-2)^2 = -1, \text{ which is absurd}$$

$$\Rightarrow f(x+2p) \neq f(x)$$

$$\text{Similarly } f(x+3p) \neq f(x)$$

Now

$$f(x+4p) = \frac{f(x+3p)-5}{f(x+3p)-3} = \frac{\frac{3f(x)-5}{f(x)-1} - 5}{\frac{3f(x)-5}{f(x)-1} - 3} = f(x)$$

$$\Rightarrow \text{Period of } f(x) \text{ is } 4p$$

W.E-6: For any real number 't', [t] denotes the G.I.F.

Let W be the set of all non-negative integers and

$f : W \rightarrow W$ is a function defined by

$$f(x) = \left(x - 10 \cdot \left[\frac{x}{10} \right] \right) \cdot 10^{\lfloor \log_{10} x \rfloor} + f\left(\left[\frac{x}{10} \right]\right); x > 0$$

$$= 0; \quad x = 0$$

if $f(7752)$ is a four digit number abcd then

$$(c+d) - (a+b) =$$

a) 5 b) 7 c) 3 d) 2

Key. b

Sol. $f(7752) = 2557 \Rightarrow c + d - (a + b) = 7$

W.E-7: Suppose $f: [-2, 2] \rightarrow R$ defined

by $f(x) = \begin{cases} -1, & \text{for } -2 \leq x < 0 \\ x-1; & \text{for } 0 \leq x \leq 2 \end{cases}$ then

$\{x \mid x \in [-2, 2]: x \leq 0 \text{ and } f(|x|) = x\} =$

a) $\{-1\}$ b) $\{0\}$ c) $\left\{\frac{-1}{2}\right\}$ d) ϕ

Key. c

Sol. $|x| \in [0, 2]$

$\therefore f(|x|) = |x| - 1 = x \Rightarrow x = -\frac{1}{2} \quad (x \leq 0)$

W.E-8: Let $f: \left(\frac{1}{2}, 1\right) \rightarrow [0, \infty)$ and

$f(x) = \sqrt{\log_e \left(\frac{x}{1-x}\right)}$ then $f(x)$ is

- a) one-one function b) Many-one function
c) Into function d) bijection

Key. c

Sol. Range is $(0, \infty)$

W.E-9: Number of positive integers in the domain of

the function $f(x) = \sqrt{\log_{0.5} \log_6 \left(\frac{x^2 + x}{x+4}\right)}$ is

Key. 6

Sol. Integers are 3, 4, 5, 6, 7, 8

W.E-10: Let $f(x)$ be a non-constant polynomial satisfying the relation

$f(x) \cdot f(y) = f(x) + f(y) + f(xy) - 2$, for all real 'x', 'y' and $f(0) \neq 1$, suppose $f(4) = 65$, then

a) $f'(x)$ is a polynomial of degree 2

b) roots of $f'(x) = 2x + 1$ are real

c) $x \cdot f'(x) = 3(f(x) - 1)$

d) $f'(-1) = 3$

Key. a, b, c, d

Sol. Given

$f(x) \cdot f(y) = f(x) + f(y) + f(xy) - 2, \forall x, y \in R$

Put $x = 1 = y$ then

$(f(1))^2 - 3f(1) + 2 = 0 \Rightarrow f(1) = 1 \text{ (or) } 2$

If $f(1) = 1$ then $f(1) = 1, \forall x \in R$. A contradiction

$\therefore$ degree of $f(x)$ is positive

$\therefore f(1) \neq 1$. Hence $f(1) = 2$

replace 'y' by ' $\frac{1}{x}$ ', then $f(x) \cdot f\left(\frac{1}{x}\right) = 2$
($\because f(1) = 2$)

$f(x) = \pm x^n + 1 \Rightarrow f(4) = 65$

$\Rightarrow f(x) = x^3 + 1 \Rightarrow f'(x) = 3x^2$

W.E-11: The least value of the function

$f(x) = \left(\frac{2 + \cos x}{\sin x}\right)$ for $0 < x < \pi$, is

a) 2 b) $\frac{1}{2}$ c) 3 d) $\frac{1}{3}$

Key. c

Sol. Put $\cos x = t$ so that $-1 < t < 1$

$f(t) = \frac{(2+t)^2}{1-t^2} \Rightarrow f'(t) = 0 \Rightarrow t = -\frac{1}{2}$

$f'(t) < 0$ for $t < -\frac{1}{2}$; $f'(t) > 0$ for $t > -\frac{1}{2}$

$\Rightarrow f(t)$ is least at $t = -\frac{1}{2}$ and $f(t) = 3$

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W.E-12: The range of the function

$$f(x) = \log_2 \left(2 - \log_{\sqrt{2}} (16 \sin^2 x + 1) \right)$$

- a) $[0, 1]$ b) $(-\infty, 1]$ c) $[-1, 1]$ d) $(-\infty, \infty)$

Key. b

Sol. Here $16 \sin^2 x + 1 > 0$ and

$$2 - \log_{\sqrt{2}} (16 \sin^2 x + 1) > 0$$

W.E-13: The range of $f: R \rightarrow R$,

$$f(x) = \frac{(\sqrt{x^2 + 1} - 3x)}{\sqrt{x^2 + 1} + x} \text{ is}$$

- a) $(0, \infty)$ b) $(-1, \infty)$
c) $(-\infty, -1)$ d) $[1, \infty)$

Key. b

Sol. Let $\sqrt{x^2 + 1} + x = t \Rightarrow \sqrt{x^2 + 1} - x = \frac{1}{t}$

$$\Rightarrow f(t) = -1 + \frac{1}{t^2} \Rightarrow \text{range} = (-1, \infty)$$

W.E-14: If $f(x) + f(y) = f(x+y)f(x-y) \forall x, y$

then $f(x)$ is

- a) even b) odd
c) Neither even nor odd
d) both even, odd

Key. a

Sol. Interchange x and y

$$\Rightarrow f(y) + f(x) = f(y+x)f(y-x)$$

$$\Rightarrow f(x-y) = f(y-x) \text{ putting } y = 2x$$

$$\Rightarrow f(x) = f(-x)$$

W.E-15: The number of roots of the equation

$$|\cos |x|| = x + |x|, |x| \leq 2\pi \text{ is}$$

- a) 0 b) 1 c) 2 d) 3

Key. d

Sol. $x < 0 \Rightarrow |\cos x| = 0 \Rightarrow x = \frac{-\pi}{2}, \frac{-3\pi}{2}; x > 0$

$$\Rightarrow \cos x = 2x \text{ has one root in } \left(0, \frac{\pi}{2}\right)$$

W.E-16: Let $f: R - \left\{\frac{3}{2}\right\} \rightarrow R, f(x) = \frac{3x+5}{2x-3}$

$$\text{Let } f_2(x) = f(f(x)); f_3(x) = f(f_2(x)); \dots, f_n(x) = f(f_{n-1}(x)), \text{ then } f_{2008}(x) + f_{2009}(x) =$$

$$\text{Sol. } f(x) = \frac{3x+5}{2x-3};$$

$$f_2(x) = f(f(x)) = \frac{3f(x)+5}{2f(x)-3} = x$$

$$\therefore f_2(x) = f_4(x) = \dots = f_{2008}(x) = x$$

$$f(x) = f_3(x) = \dots = f_{2009}(x) = \frac{3x+5}{2x-3}$$

$$\therefore f_{2008}(x) + f_{2009}(x) = x + \frac{3x+5}{2x-3} = \frac{2x^2+5}{2x-3}$$